

END-TO-END TEST REPORT

Mobile Application, Admin Panel and Backend API

Project	SRV Electricals Digital Platform
Test date	18 June 2026
Environment	Local development services connected to the project database
Scope	Mobile app, admin UI, backend API, database integrity and Android build
Prepared for	Management and deployment review
Overall verdict	CONDITIONAL GO - functional suite passes; inventory data requires remediation

129	124	1	4
TESTS	PASSED	FAILED	BLOCKED

1. Executive Summary

The tested platform is functionally stable across the mobile application, administrator interface and backend API. All builds and static checks passed. The automated regression suite completed 102 API and data checks with 101 passes and one genuine failure. Eleven administrator UI checks and six focused multiple-photo checks also passed.

Release recommendation: Proceed to controlled device acceptance testing after inventory stock values are corrected. Do not treat the current database as production-ready for product ordering.

Implemented feature: Need Help now supports up to five photos with a visible plus button, multi-selection, previews, removal and ticket-history display. Enquiry Support receives and displays every image in a lazy-loaded gallery.

2. Scope and Test Approach

- Backend contract and authorization checks against the running NestJS API.
- Role-based mobile API checks for electrician, dealer, customer and counter boy identities.
- Data-integrity checks for products, banners, QR inventory and product orders.
- Browser-based administrator regression for support, QR, products, delivery, notifications, settings and electrician filters.
- Production/static compilation checks for backend, admin, mobile TypeScript, lint and Android native code.

- Focused realistic multi-photo payload test, database verification and administrator gallery verification.

3. Feature Verification - Multiple Support Photos

Layer	Verified behaviour	Result
Mobile UI	Visible + button, 0/5 counter, multiple selection, all-photo confirmation preview and individual removal.	PASS
Mobile transport	Sends photoUrls array and legacy photoUrl for backward compatibility.	PASS
Backend	Accepts and stores up to five unique photo values in a PostgreSQL text array.	PASS
Realistic API payload	Three approximately 260 KB images submitted successfully with HTTP 201.	PASS
Database	cardinality(photoUrls) verified as 3 before test fixture cleanup.	PASS
Admin API	Ticket detail returned all three photo values.	PASS
Admin UI	Attachment badge displayed 3 photos; detail drawer rendered three clickable images.	PASS

4. Results by Area

Area	Outcome	Assessment
Backend API	101 / 102 passed	One inventory data failure; all exercised endpoints responded correctly.
Mobile role APIs	48 / 48 passed	Authentication and core role services passed for all four roles.
Admin safe GET APIs	30 / 30 passed	All parameter-free administrator read endpoints in the OpenAPI contract passed.
Admin browser regression	11 / 11 passed	Targeted screens loaded with live API data and no visible error alerts.
Build and static analysis	6 / 6 passed	Backend, admin, mobile lint/type checks and Android build passed.
Physical acceptance	0 / 3 executed	Requires a connected handset and external payment/push services.

5. Defect and Risk Register

ID	Severity	Area	Finding / impact	Recommended action
BUG-001	HIGH	Inventory	272 of 273 active products have stock <= 0. Ordering can incorrectly report products as unavailable.	Load correct stock or agree and implement a non-stock-controlled ordering policy before deployment.

ID	Severity	Area	Finding / impact	Recommended action
QA-001	MEDIUM	Backend tests	Jest has no unit/integration test files. Regression protection currently depends on external smoke testing.	Add service, controller and payment/support integration tests to CI.
DATA-001	MEDIUM	Categories	Dedicated product_categories table contains 0 rows although the mobile API derives 30 categories from products.	Confirm intended source of truth and populate/synchronize the category table before production administration.
ENV-001	MEDIUM	Push	No physical device token was available, so OS-level notification receipt was not executed.	Install the current native build, log in, confirm token registration, then send role and individual test notifications.
ENV-002	MEDIUM	Payment	Razorpay native compilation passed, but UPI handoff and callback require physical-device acceptance.	Run success, cancel and failure flows using Razorpay test mode on a connected handset.
TECH-001	LOW	Android build	Gradle reports deprecated features that will be incompatible with Gradle 9.0.	Record dependency upgrades in the next maintenance sprint; no current build failure.

6. Deployment Acceptance Checklist

- **PASS:** Backend and admin production builds complete without compilation errors.
- **PASS:** Mobile TypeScript, Expo lint, admin lint and native Android compilation pass.
- **PASS:** Multiple support photos persist through app -> API -> database -> admin gallery.
- **ACTION:** Correct active product stock data and retest order creation.
- **ACTION:** Perform connected-device push, image picker and Razorpay acceptance tests.
- **ACTION:** Decide and document the production source of truth for product categories.

Appendix A - Complete Test Case Register

This appendix lists every executed, failed and blocked test included in the reported totals. Temporary test tickets and the temporary test administrator were removed after verification.

ID	Module	Test case	Result	Evidence
TC-001	Platform	Backend health endpoint	PASS	HTTP 200 (126 ms)
TC-002	Platform	OpenAPI specification available	PASS	HTTP 200 (34 ms)
TC-003	Catalog	Active product catalog	PASS	HTTP 200 (51 ms)
TC-004	Catalog	Product category catalog	PASS	HTTP 200 (17 ms)
TC-005	Content	Mobile banners	PASS	HTTP 200 (15 ms)
TC-006	Settings	Maintenance configuration	PASS	HTTP 200 (10 ms)
TC-007	Settings	Application settings	PASS	HTTP 200 (15 ms)
TC-008	Marketing	Offers feed	PASS	HTTP 200 (9 ms)
TC-009	Content	Testimonials feed	PASS	HTTP 200 (11 ms)

ID	Module	Test case	Result	Evidence
TC-010	Rewards	Gift product catalog	PASS	HTTP 200 (9 ms)
TC-011	Rewards	Reward schemes	PASS	HTTP 200 (8 ms)
TC-012	Content	Festival theme endpoint	PASS	HTTP 200 (8 ms)
TC-013	Authentication	electrician profile authorization	PASS	HTTP 200 (33 ms)
TC-014	Notifications	electrician notification inbox	PASS	HTTP 200 (11 ms)
TC-015	Support	electrician support ticket history	PASS	HTTP 200 (222 ms)
TC-016	Orders	electrician product order history	PASS	HTTP 200 (46 ms)
TC-017	Profile	electrician profile QR	PASS	HTTP 200 (38 ms)
TC-018	Wallet	electrician wallet summary	PASS	HTTP 200 (40 ms)
TC-019	QR & Rewards	electrician scan history	PASS	HTTP 200 (17 ms)
TC-020	Rewards	electrician redemption history	PASS	HTTP 200 (8 ms)
TC-021	Referral	electrician referral details	PASS	HTTP 200 (8 ms)
TC-022	Feedback	electrician app rating endpoint	PASS	HTTP 200 (8 ms)
TC-023	Media	electrician plays feed	PASS	HTTP 200 (9 ms)
TC-024	Cart	electrician cart endpoint	PASS	HTTP 200 (11 ms)
TC-025	Authentication	dealer profile authorization	PASS	HTTP 200 (21 ms)
TC-026	Notifications	dealer notification inbox	PASS	HTTP 200 (11 ms)
TC-027	Support	dealer support ticket history	PASS	HTTP 200 (10 ms)
TC-028	Orders	dealer product order history	PASS	HTTP 200 (13 ms)
TC-029	Profile	dealer profile QR	PASS	HTTP 200 (18 ms)
TC-030	Wallet	dealer wallet summary	PASS	HTTP 200 (43 ms)
TC-031	QR & Rewards	dealer scan history	PASS	HTTP 200 (19 ms)
TC-032	Rewards	dealer redemption history	PASS	HTTP 200 (11 ms)
TC-033	Referral	dealer referral details	PASS	HTTP 200 (10 ms)
TC-034	Feedback	dealer app rating endpoint	PASS	HTTP 200 (9 ms)
TC-035	Media	dealer plays feed	PASS	HTTP 200 (10 ms)
TC-036	Cart	dealer cart endpoint	PASS	HTTP 200 (10 ms)
TC-037	Authentication	user profile authorization	PASS	HTTP 200 (23 ms)
TC-038	Notifications	user notification inbox	PASS	HTTP 200 (11 ms)
TC-039	Support	user support ticket history	PASS	HTTP 200 (9 ms)
TC-040	Orders	user product order history	PASS	HTTP 200 (10 ms)
TC-041	Profile	user profile QR	PASS	HTTP 200 (19 ms)
TC-042	Wallet	user wallet summary	PASS	HTTP 200 (37 ms)
TC-043	QR & Rewards	user scan history	PASS	HTTP 200 (15 ms)
TC-044	Rewards	user redemption history	PASS	HTTP 200 (11 ms)
TC-045	Referral	user referral details	PASS	HTTP 200 (9 ms)

ID	Module	Test case	Result	Evidence
TC-046	Feedback	user app rating endpoint	PASS	HTTP 200 (8 ms)
TC-047	Media	user plays feed	PASS	HTTP 200 (9 ms)
TC-048	Cart	user cart endpoint	PASS	HTTP 200 (8 ms)
TC-049	Authentication	counterboy profile authorization	PASS	HTTP 200 (18 ms)
TC-050	Notifications	counterboy notification inbox	PASS	HTTP 200 (12 ms)
TC-051	Support	counterboy support ticket history	PASS	HTTP 200 (10 ms)
TC-052	Orders	counterboy product order history	PASS	HTTP 200 (10 ms)
TC-053	Profile	counterboy profile QR	PASS	HTTP 200 (21 ms)
TC-054	Wallet	counterboy wallet summary	PASS	HTTP 200 (36 ms)
TC-055	QR & Rewards	counterboy scan history	PASS	HTTP 200 (17 ms)
TC-056	Rewards	counterboy redemption history	PASS	HTTP 200 (8 ms)
TC-057	Referral	counterboy referral details	PASS	HTTP 200 (10 ms)
TC-058	Feedback	counterboy app rating endpoint	PASS	HTTP 200 (7 ms)
TC-059	Media	counterboy plays feed	PASS	HTTP 200 (10 ms)
TC-060	Cart	counterboy cart endpoint	PASS	HTTP 200 (9 ms)
TC-061	Admin Authentication	Admin profile authorization	PASS	HTTP 200 (10 ms)
TC-062	Support	Create support query with three realistic-size photos	PASS	HTTP 201 (145 ms)
TC-063	Support	Database preserves every uploaded support photo	PASS	Stored photo count: 3
TC-064	Support	Admin support API returns all ticket photos	PASS	HTTP 200 (200 ms)
TC-065	Admin API Regression	Get current admin profile	PASS	HTTP 200 (12 ms)
TC-066	Admin API Regression	Get all distinct electrician states for filter dropdown	PASS	HTTP 200 (16 ms)
TC-067	Admin API Regression	Get all distinct electrician subCategories for filter dropdown	PASS	HTTP 200 (13 ms)
TC-068	Admin API Regression	Get electrician tier distribution counts	PASS	HTTP 200 (12 ms)
TC-069	Admin API Regression	Get all distinct dealer states for filter dropdown	PASS	HTTP 200 (13 ms)
TC-070	Admin API Regression	Get dealer stats (total, active, pending counts)	PASS	HTTP 200 (12 ms)
TC-071	Admin API Regression	Get QR code stats (total, active, used)	PASS	HTTP 200 (16 ms)
TC-072	Admin API Regression	Get scan statistics	PASS	HTTP 200 (87 ms)
TC-073	Admin API Regression	Get all notifications	PASS	HTTP 200 (16 ms)
TC-074	Admin API Regression	Get all banners	PASS	HTTP 200 (14 ms)

ID	Module	Test case	Result	Evidence
TC-075	Admin API Regression	Get referral statistics	PASS	HTTP 200 (43 ms)
TC-076	Admin API Regression	Get financial summary	PASS	HTTP 200 (53 ms)
TC-077	Admin API Regression	Get all financial transactions	PASS	HTTP 200 (53 ms)
TC-078	Admin API Regression	Get dealer bonus summary	PASS	HTTP 200 (28 ms)
TC-079	Admin API Regression	Get points transfer history	PASS	HTTP 200 (189 ms)
TC-080	Admin API Regression	Get dashboard analytics	PASS	HTTP 200 (27 ms)
TC-081	Admin API Regression	Get scan analytics	PASS	HTTP 200 (126 ms)
TC-082	Admin API Regression	Get user analytics	PASS	HTTP 200 (69 ms)
TC-083	Admin API Regression	Get revenue analytics	PASS	HTTP 200 (73 ms)
TC-084	Admin API Regression	Get all support tickets	PASS	HTTP 200 (1066 ms)
TC-085	Admin API Regression	Get all settings	PASS	HTTP 200 (16 ms)
TC-086	Admin API Regression	Get all product categories	PASS	HTTP 200 (14 ms)
TC-087	Admin API Regression	Get app user states	PASS	HTTP 200 (13 ms)
TC-088	Admin API Regression	Get app user stats	PASS	HTTP 200 (12 ms)
TC-089	Admin API Regression	Get counter boy states	PASS	HTTP 200 (10 ms)
TC-090	Admin API Regression	Get counter boy stats	PASS	HTTP 200 (11 ms)
TC-091	Admin API Regression	Get play stats	PASS	HTTP 200 (10 ms)
TC-092	Admin API Regression	Get all app icons	PASS	HTTP 200 (14 ms)
TC-093	Admin API Regression	Get currently active app icon	PASS	HTTP 200 (15 ms)
TC-094	Admin API Regression	Get product order stats	PASS	HTTP 200 (70 ms)
TC-095	Catalog Data Integrity	Active products available	PASS	Database count: 273
TC-096	Catalog Data Integrity	Active products have names	PASS	Database count: 0
TC-097	Catalog Data Integrity	Active products have unique SKU values	PASS	Database count: 0

ID	Module	Test case	Result	Evidence
TC-098	Catalog Data Integrity	Active products have valid nonnegative prices	PASS	Database count: 0
TC-099	Catalog Data Integrity	Active products are in stock	FAIL	Database count: 272
TC-100	Content Data Integrity	Active banners have image references	PASS	Database count: 0
TC-101	QR & Rewards Data Integrity	QR inventory exists	PASS	Database count: 11710
TC-102	Orders Data Integrity	Product orders exist for workflow validation	PASS	Database count: 13
BLD-001	Backend Build	NestJS production compilation	PASS	npm run build completed successfully.
BLD-002	Admin Build	Next.js production compilation and TypeScript	PASS	Production build and static generation completed successfully.
BLD-003	Mobile Static	Mobile TypeScript validation	PASS	npm run tsc --noEmit completed with zero errors.
BLD-004	Mobile Static	Expo ESLint validation	PASS	npm run expo lint completed with zero errors.
BLD-005	Admin Static	Admin ESLint validation	PASS	npm run lint completed with zero errors.
BLD-006	Android Build	Android debug APK compilation	PASS	Gradle assembleDebug completed successfully; Razorpay and Expo notifications native modules compiled.
BLD-007	Backend Unit Tests	Repository Jest suite	BLOCKED	No *.spec.ts files exist; Jest returned 'No tests found'. API regression coverage was supplied separately.
UI-001	Admin UI	Administrator authentication and dashboard load	PASS	Temporary isolated test administrator authenticated; dashboard rendered without UI alerts.
UI-002	Admin UI	Enquiry Support list and filters	PASS	Ticket list, status filter, category filter and attachment counts rendered.
UI-003	Admin UI	Support ticket multiple-photo gallery	PASS	A three-photo fixture displayed USER ATTACHMENTS (3) with three individually clickable images.
UI-004	Admin UI	QR Generator quick statistics	PASS	Total 11,710; scanned 522; active 11,188 rendered from API data.
UI-005	Admin UI	QR Generator product search dropdown	PASS	Clicking search loaded the active product list with SKU, category and points.
UI-006	Admin UI	Product management page	PASS	Products page and product search control rendered with no UI alert.
UI-007	Admin UI	Delivery Tracker workflow screen	PASS	Pending, Payment Done, Dispatched, Delivered and Rejected states rendered.
UI-008	Admin UI	Notification management screen	PASS	Notification page and audience targeting controls rendered without UI alerts.
UI-009	Admin UI	App Settings push notification controls	PASS	Push notification controls rendered without UI alerts.
UI-010	Admin UI	Electrician bonus and type filters	PASS	Welcome Back Bonus, All Electrician and App Status filters rendered.
UI-011	Admin UI	Electrician API data population	PASS	Loading completed with 50 visible table rows and no load error.

ID	Module	Test case	Result	Evidence
MOB-001	Mobile Support	Visible plus-button for multiple photos	PASS	Need Help shows an explicit + control, count and multiple-photo guidance.
MOB-002	Mobile Support	Multi-select gallery configuration	PASS	Image picker enables multiple selection and limits remaining capacity.
MOB-003	Mobile Support	Five-photo maximum	PASS	Client and server both enforce a maximum of five attachments.
MOB-004	Mobile Support	Selected-photo preview and removal	PASS	All selected photos are previewed; each attached thumbnail has a remove control.
MOB-005	Mobile Support	Ticket detail attachment rendering	PASS	Submitted ticket detail displays all retained attachment URLs.
MOB-006	Admin Support	Backward-compatible photo ingestion	PASS	Admin merges legacy photoUrl and new photoUrls without duplicates.
PHY-001	Physical Device	Gallery multi-select on a real Android handset	BLOCKED	No Android handset was connected during this run; source, API and native build checks passed.
PHY-002	Physical Device	OS push notification delivery	BLOCKED	No registered connected device token was available for physical receipt verification.
PHY-003	Payment	Complete Razorpay UPI payment on handset	BLOCKED	Native Razorpay module compiled, but external UPI handoff requires a connected handset and Razorpay test transaction.

Appendix B - Test Data Snapshot

Dataset	Count	Observation
Products	274	273 active; 272 active with zero stock
Derived product categories	30	Returned by mobile product-category API
Dedicated category records	0	Database product_categories table
Active banners	4	All active banner API checks passed
QR codes	11,710	522 scanned; 11,188 remaining

Sign-off Statement

The evidence supports a conditional deployment decision. Application functionality, API integration and the newly requested multiple-photo support passed the available automated and browser checks. Production release should follow only after the inventory data issue and the three physical-device acceptance items are closed or formally accepted by the release owner.